The triangle inequality

ANSWER KEY

Core Concepts: The triangle inequality

1 In $\triangle ABC$ studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 91° ✓

b. 101°

c. 83°

d. 136°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (44^\circ + 45^\circ) = 91^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm ✓

c. 24 cm

d. 32 cm

Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 14

c. 12 ✓

d. 17

Substituting $x = 1$ into $y = 5x + 7$ gives $y = 5(1) + 7 = 12$.

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 84° ✓

b. 94°

c. 76°

d. 136°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (44^\circ + 52^\circ) = 84^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length?

a. 21 cm

b. 24 cm ✓

c. 32 cm

d. 72 cm

Multiplying the original length by the scale factor gives $8 \times 3 = 24$ cm.

6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 12

b. 19

c. 17 ✓

d. 22

Substituting $x = 2$ into $y = 5x + 7$ gives $y = 5(2) + 7 = 17$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 77° ✓

b. 87°

c. 69°

d. 136°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (44^\circ + 59^\circ) = 77^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm ✓

c. 40 cm

d. 128 cm

Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?

a. 17

b. 24

c. 22 ✓

d. 27

Substituting $x = 3$ into $y = 5x + 7$ gives $y = 5(3) + 7 = 22$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 70° ✓

b. 80°

c. 62°

d. 136°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (44^\circ + 66^\circ) = 70^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm ✓

c. 48 cm

d. 200 cm

Multiplying the original length by the scale factor gives $8 \times 5 = 40$ cm.

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 22

b. 29

c. 27 ✓

d. 32

Substituting $x = 4$ into $y = 5x + 7$ gives $y = 5(4) + 7 = 27$.

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